A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
    leftSum = -∞
    sum = 0
    for i = mid downto low
        sum = sum + A[i]
        if sum > leftSum
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

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leftSum = -∞
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            rightSum = sum
    return leftSum+rightSum
```

leftSum = -∞
sum = -2

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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    sum = 0
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        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = -2
sum = -2

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
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    return leftSum+rightSum
```

leftSum = -2
sum = -3

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

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    return leftSum+rightSum
```

leftSum = -2
sum = -3

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

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    sum = 0
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        sum = sum + A[i]
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            rightSum = sum
    return leftSum+rightSum
```

leftSum = -2
sum = 2

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
sum = 2

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
sum = -1

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
sum = -1

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
sum = 1

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
sum = 1

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = -∞
sum = 0

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = -∞
sum = 0

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
    leftSum = -∞
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    for i = mid downto low
        sum = sum + A[i]
        if sum > leftSum
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = -∞
sum = -4

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
    leftSum = -∞
    sum = 0
    for i = mid downto low
        sum = sum + A[i]
        if sum > leftSum
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = -4
sum = -4

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
    leftSum = -∞
    sum = 0
    for i = mid downto low
        sum = sum + A[i]
        if sum > leftSum
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = -4
sum = 6

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
    leftSum = -∞
    sum = 0
    for i = mid downto low
        sum = sum + A[i]
        if sum > leftSum
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 6
sum = 6

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
    leftSum = -∞
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    for i = mid downto low
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        if sum > leftSum
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 6
sum = 13

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
    leftSum = -∞
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    for i = mid downto low
        sum = sum + A[i]
        if sum > leftSum
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 13
sum = 13

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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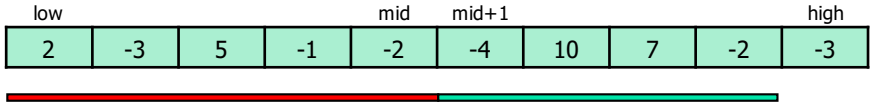
            leftSum = sum
    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 13
sum = 11



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    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 13
sum = 11

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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        if sum > leftSum
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    rightSum = -∞
    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 13
sum = 8

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
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    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 13
sum = 8

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

```
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    sum = 0
    for i = mid+1 to high
        sum = sum + A[i]
        if sum > rightSum
            rightSum = sum
    return leftSum+rightSum
```

leftSum = 2
rightSum = 13
sum = 8

return 15

low				mid	mid+1				high
2	-3	5	-1	-2	-4	10	7	-2	-3

A Divide-and-Conquer Algorithm

- 앞에서 만든 FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high) 함수를 이용해서 FIND-MAXIMUM-SUBARRAY3(A, low, high)를 divide and conqer로 작성해 보세요.
  - maximum range의 array 상의 왼쪽인덱스, 오른쪽 인덱스, maximum range sum 이 세가지를 리턴하세요.

A Divide-and-Conquer Algorithm

```
FIND-MAXIMUM-SUBARRAY3(A, low, high)
    if high == low
        return (low, high, A[low] )
    else  mid = (low +high)/2
        (leftLow, leftHigh, leftSum) =  FIND-MAXIMUM-SUBARRAY3(A, low, mid)
        (rightLow, rightHigh, rightSum) = FIND-MAXIMUM-SUBARRAY3(A, mid+1, high)
        (crossLow, crossHigh, crossSum) = FIND-MAX-CROSSING-SUBARRAY(A, low, mid, high)
        if leftSum >= rightSum and leftSum >= crossSum
            return (leftLow, leftHigh, leftSum)
        if rightSum >= leftSum and rightSum >= crossSum
            return (rightLow, rightHigh, rightSum)
        else return (crossLow, crossHigh, crossSum)
```

Incremental Algorithm

- Kadane's Algorithm
- If we know
  - the maximum subarray sum ending at position A[i] (Call it ThisSum)
  - the maximum subarray sum for the range [1, i] (Call it MaxSum)
- What is the maximum subarray sum for the range [i, i+1]?
  - max(MaxSum, A[i+1], ThisSum + A[i+1])